1. ATM Transaction Sequence Simulation

PYTHON ACTIVITY-BASED LEARNING | CASE STUDY

PROBLEM

Simulate a simple ATM withdrawal and show the remaining balance.

OBJECTIVE

Practise input, arithmetic, and displaying results.

PYTHON CODE

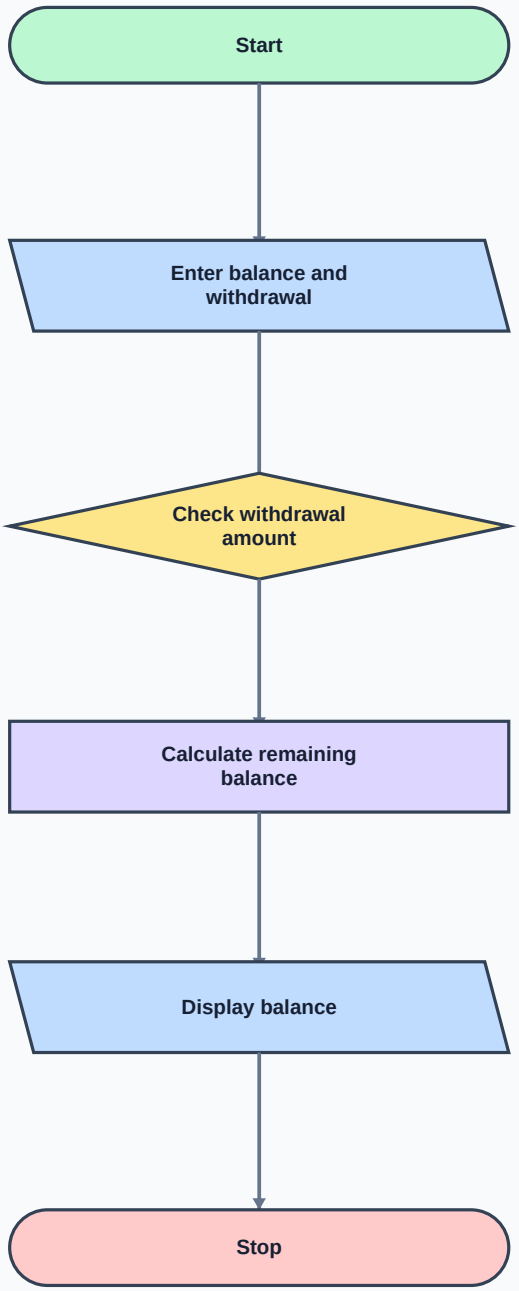
```
balance = 5000
withdraw = 1200
balance = balance - withdraw
print('Balance:', balance)
```

OUTPUT SCREEN

```
ATM TRANSACTION
Starting balance: Rs. 5000
Withdrawn: Rs. 1200
Remaining balance: Rs. 3800
```

VERTICAL FLOWCHART

Follow the arrows from top to bottom



CONCLUSION The program subtracts the withdrawal amount from the balance.

2. Student Performance Trend Analysis

PYTHON ACTIVITY-BASED LEARNING | CASE STUDY

PROBLEM

Calculate a student's average marks in three subjects.

OBJECTIVE

Practise lists, sum, and average calculation.

PYTHON CODE

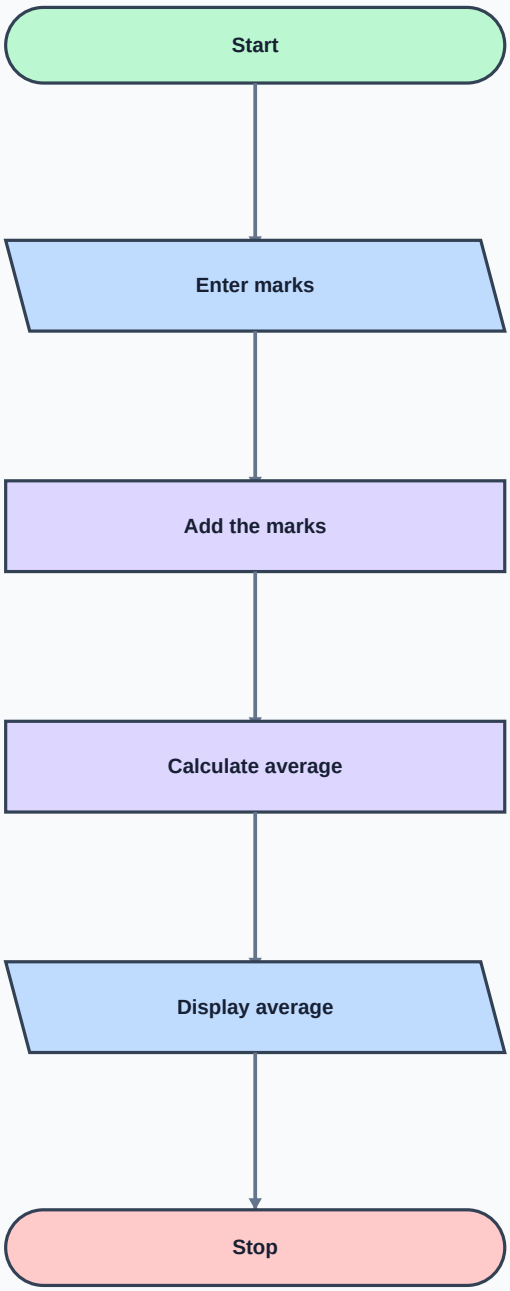
```
marks = [78, 85, 90]
total = sum(marks)
average = total / len(marks)
print('Average:', average)
```

OUTPUT SCREEN

```
STUDENT PERFORMANCE
Marks: [78, 85, 90]
Total: 253
Average: 84.3333333333
```

VERTICAL FLOWCHART

Follow the arrows from top to bottom



CONCLUSION The program calculates the average of the entered marks.

3. Advanced Numeric Pattern Generator

PYTHON ACTIVITY-BASED LEARNING | CASE STUDY

PROBLEM

Print the squares of numbers from 1 to 5.

OBJECTIVE

Practise loops and numeric calculations.

PYTHON CODE

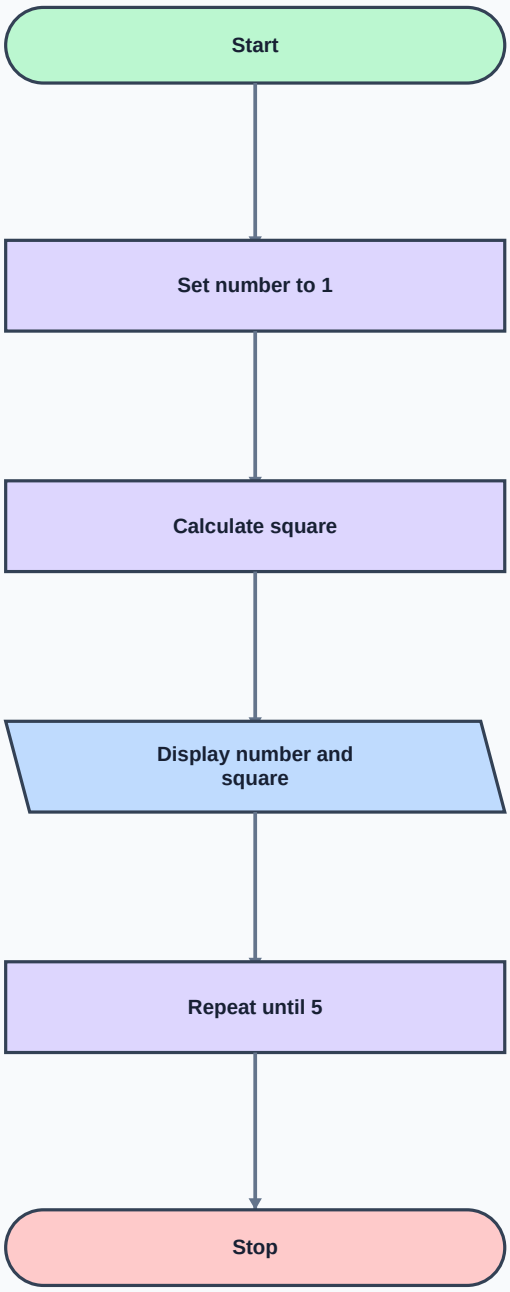
```
for n in range(1, 6):
    square = n * n
    print(n, '->', square)
```

OUTPUT SCREEN

```
NUMBER SQUARES
1 -> 1
2 -> 4
3 -> 9
4 -> 16
5 -> 25
```

VERTICAL FLOWCHART

Follow the arrows from top to bottom



CONCLUSION The loop prints each number and its square.

4. Password Policy Validation Engine

PYTHON ACTIVITY-BASED LEARNING | CASE STUDY

PROBLEM

Check whether a password has at least eight characters.

OBJECTIVE

Practise string length and if-else conditions.

PYTHON CODE

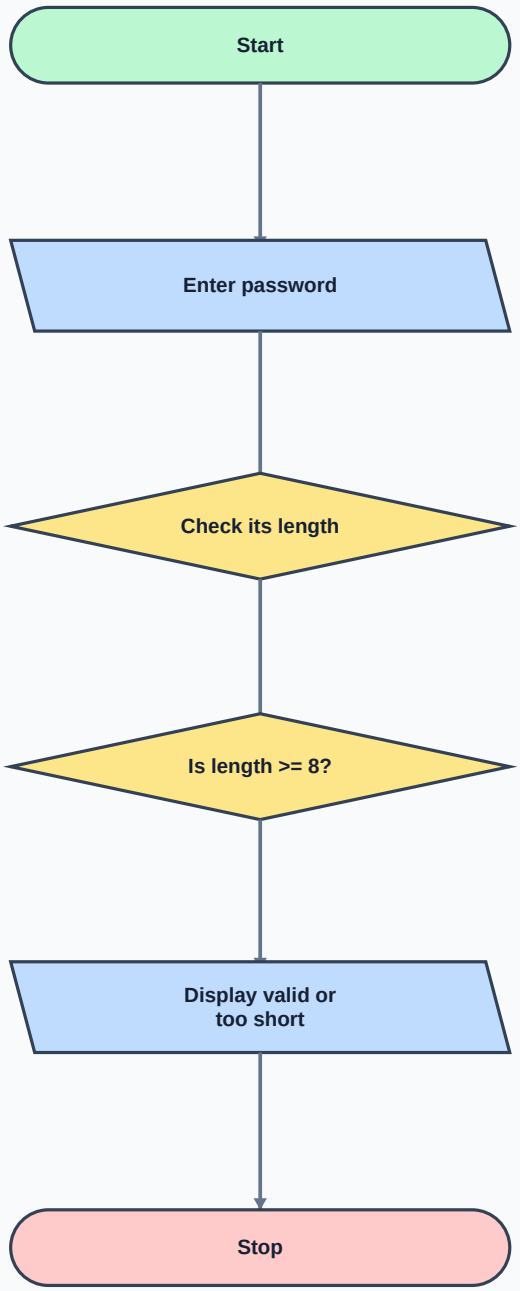
```
password = 'Study@123'
if len(password) >= 8:
    print('Password accepted')
else: print('Too short')
```

OUTPUT SCREEN

```
PASSWORD CHECK
Password entered: Study@123
Password accepted
```

VERTICAL FLOWCHART

Follow the arrows from top to bottom



CONCLUSION The program accepts passwords with eight or more characters.

5. Text Compression Using Character Runs

5/10

PYTHON ACTIVITY-BASED LEARNING | CASE STUDY

PROBLEM

Count how many times each character appears in a short word.

OBJECTIVE

Practise dictionaries and character counting.

PYTHON CODE

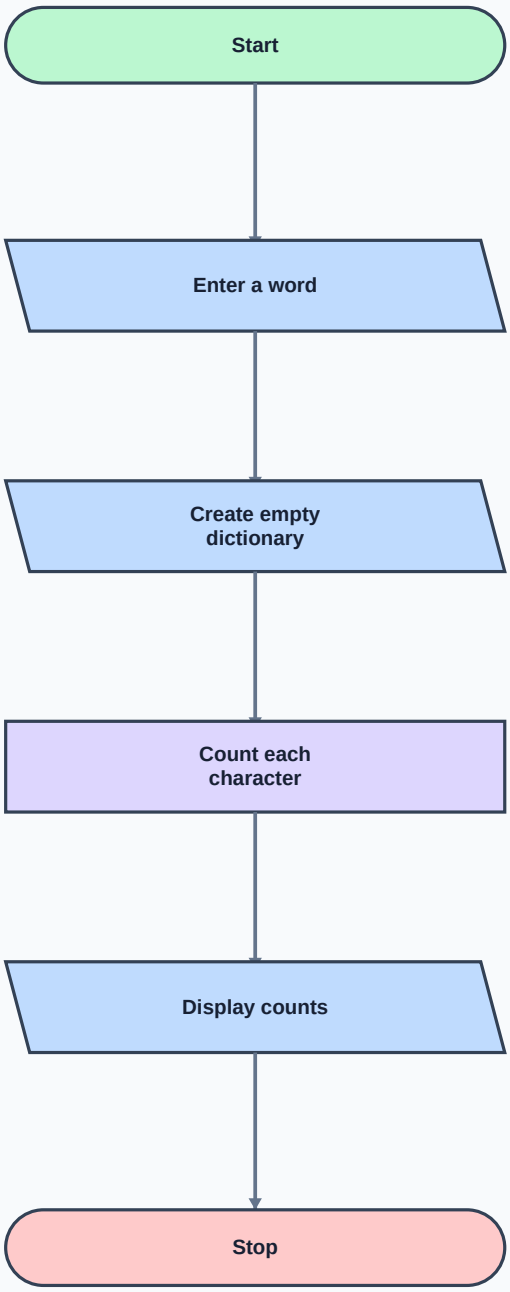
```
word = 'banana'
counts = {}
for ch in word: counts[ch] = counts.get(ch, 0) + 1
print(counts)
```

OUTPUT SCREEN

```
CHARACTER COUNTS
Word: banana
{'b': 1, 'a': 3, 'n': 2}
```

VERTICAL FLOWCHART

Follow the arrows from top to bottom



CONCLUSION The dictionary stores the number of times each character appears.

6. Student Marks Matrix Analysis

PYTHON ACTIVITY-BASED LEARNING | CASE STUDY

PROBLEM

Store marks for two students and find each student's total.

OBJECTIVE

Practise nested lists and sum.

PYTHON CODE

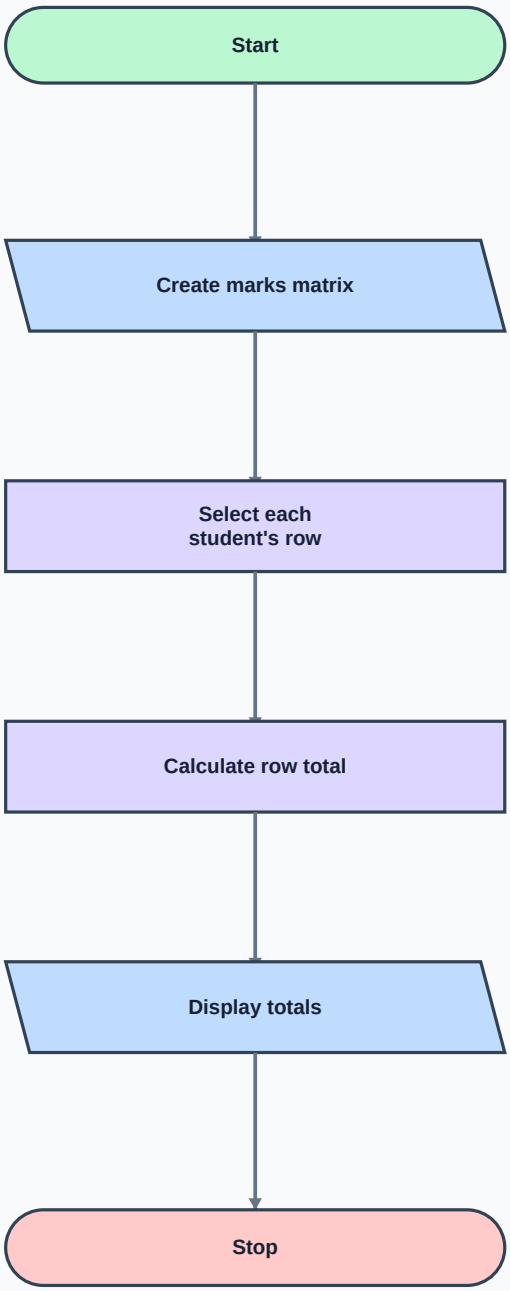
```
marks = [[80, 75, 90], [70, 85, 88]]
for row in marks:
    print('Total:', sum(row))
```

OUTPUT SCREEN

```
MARKS MATRIX
Student 1 marks: 80, 75, 90
Student 1 total: 245
Student 2 marks: 70, 85, 88
Student 2 total: 243
```

VERTICAL FLOWCHART

Follow the arrows from top to bottom



CONCLUSION The program calculates totals for each row in the marks matrix.

7. Tuple-Based Product Catalogue Audit

PYTHON ACTIVITY-BASED LEARNING | CASE STUDY

PROBLEM

Display a product's name and price stored in a tuple.

OBJECTIVE

Practise tuples and indexing.

PYTHON CODE

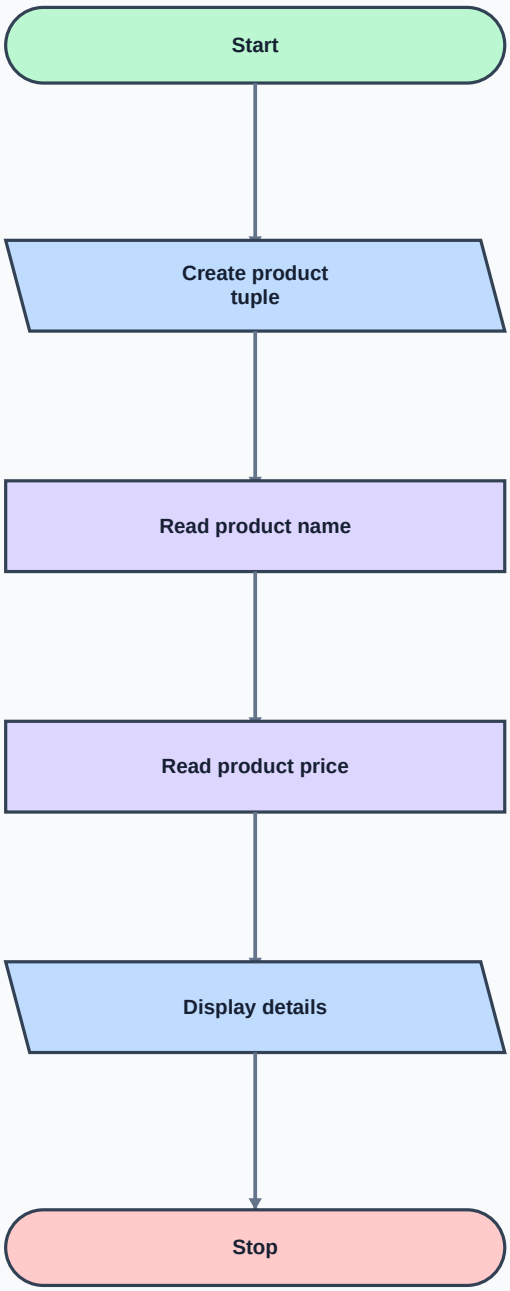
```
product = ('Notebook', 60)
name = product[0]
price = product[1]
print(name, 'Rs.', price)
```

OUTPUT SCREEN

```
PRODUCT CATALOGUE
Product: Notebook
Price: Rs. 60
```

VERTICAL FLOWCHART

Follow the arrows from top to bottom



CONCLUSION The tuple keeps product details together and the program displays them.

8. Food Waste Monitoring System

PYTHON ACTIVITY-BASED LEARNING | CASE STUDY

PROBLEM

Calculate the amount of food left after serving students.

OBJECTIVE

Practise subtraction and basic input data.

PYTHON CODE

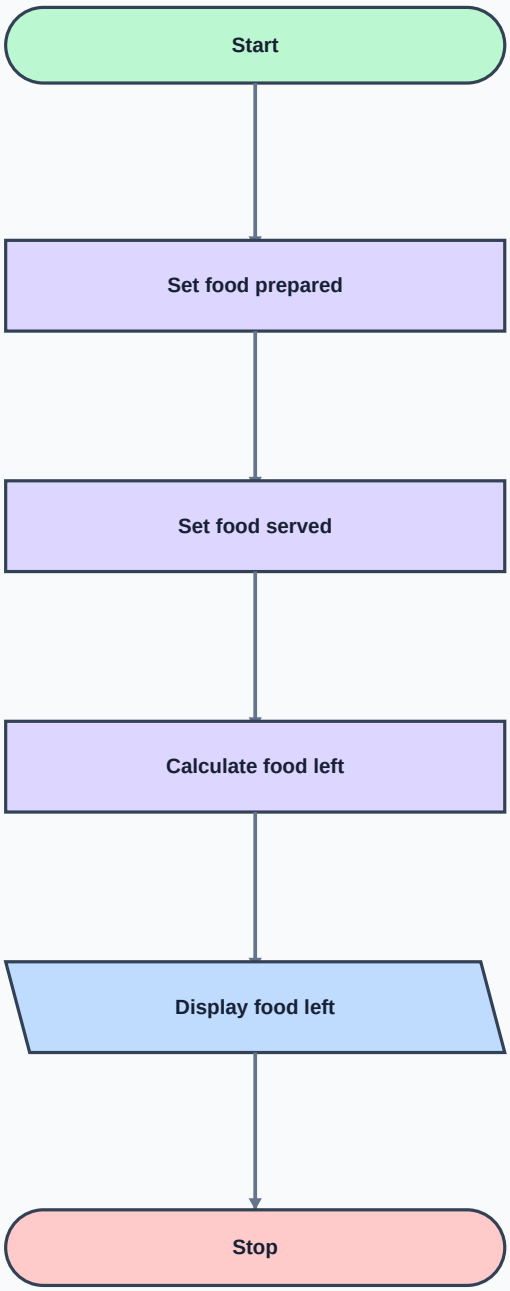
```
prepared = 50
served = 42
left = prepared - served
print('Food left:', left, 'plates')
```

OUTPUT SCREEN

```
FOOD WASTE MONITOR
Food prepared: 50 plates
Food served: 42 plates
Food left: 8 plates
```

VERTICAL FLOWCHART

Follow the arrows from top to bottom



CONCLUSION The program calculates how many plates of food remain.

9. Multi-Product Price Fluctuation Analysis

PYTHON ACTIVITY-BASED LEARNING | CASE STUDY

PROBLEM

Find the price difference between an old and a new price.

OBJECTIVE

Practise arithmetic and positive/negative differences.

PYTHON CODE

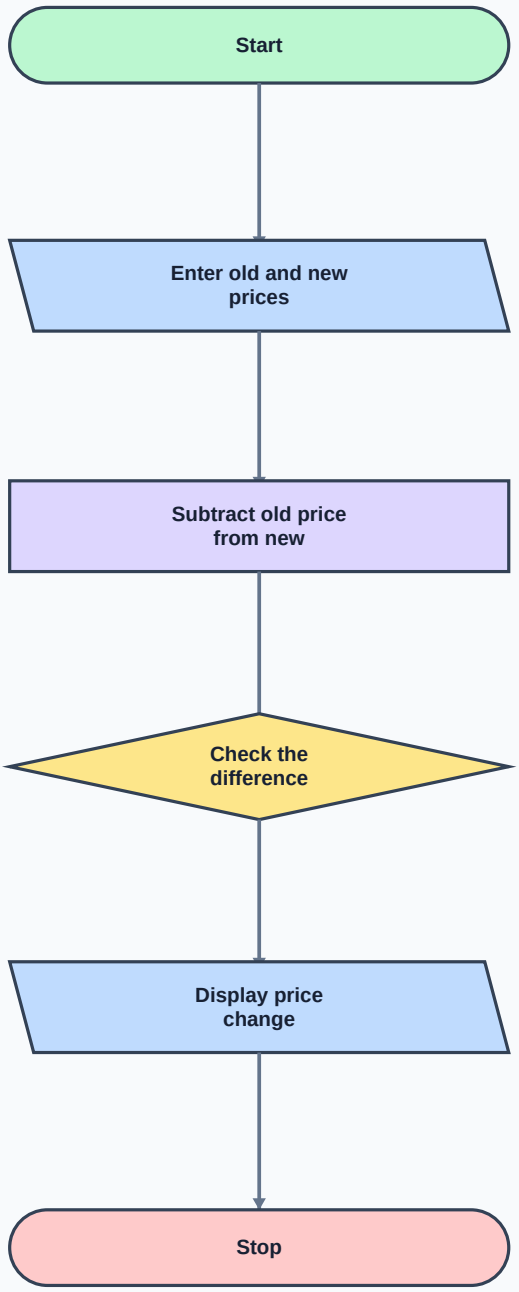
```
old_price = 100
new_price = 115
change = new_price - old_price
print('Price change:', change)
```

OUTPUT SCREEN

```
PRICE CHANGE REPORT
Old price: Rs. 100
New price: Rs. 115
Price change: Rs. 15 increase
```

VERTICAL FLOWCHART

Follow the arrows from top to bottom



CONCLUSION The program shows whether the price increased or decreased.

10. Smart Canteen Order and Stock Management

10/10

PYTHON ACTIVITY-BASED LEARNING | CASE STUDY

PROBLEM

Check whether enough stock is available for a canteen order.

OBJECTIVE

Practise comparisons and conditional statements.

PYTHON CODE

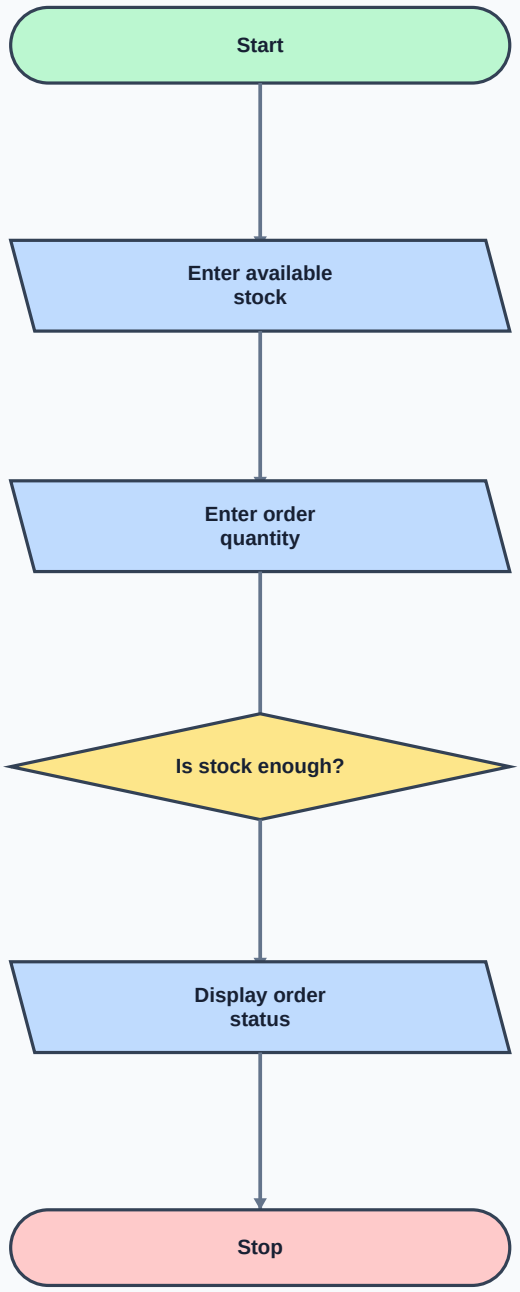
```
stock = 20
order = 6
if order <= stock: print('Order accepted')
else: print('Not enough stock')
```

OUTPUT SCREEN

```
SMART CANTEEN
Available stock: 20
Order quantity: 6
Order accepted
Remaining stock after order: 14
```

VERTICAL FLOWCHART

Follow the arrows from top to bottom



CONCLUSION The program checks stock availability before accepting an order.